AUTHORIZATION AND PAYMENT INFRASTRUCTURE FOR AGENTIC SYSTEMS

Kaldrix

The approval desk for the age of AI agents.

Every action checked. Every budget protected. Every decision remembered.

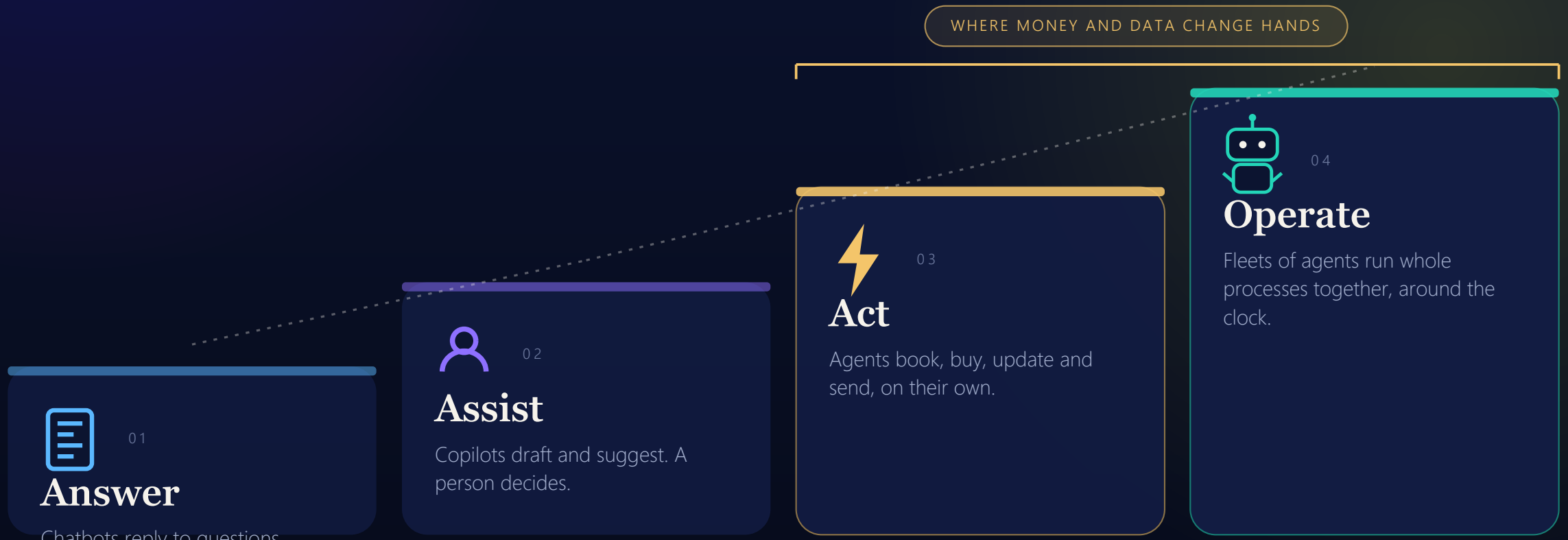
KALDRIX DECIDES. ZENTRIX MOVES THE MONEY.

CHECKED PROTECTED REMEMBERED



THE SHIFT

AI has stopped answering. It now acts.



THE PROBLEM IN ONE PICTURE

Two helpers. One account. Nobody in charge.

I will spend $40,000.



AGENT A

Looked at $50,000 a moment ago.



COMPANY BUDGET

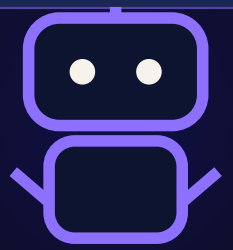
$50,000

$50,000 – $40,000 – $40,000

–$30,000

OVERSPENT

Me too: $40,000.



AGENT B

Looked at $50,000 a moment ago.

IN PLAIN WORDS

Each helper made a sensible choice from what it saw. Together they spent money that was never there.

UNDER THE HOOD

A stale read plus a non-atomic budget check: the classic check-then-act race, now with an agent holding real credentials.

THE ANSWER

Put one desk in the middle.



AGENT A · APPROVED

$40,000 spent

$10,000 left in the budget.

AGENT B · REFUSED

Stopped, with a reason

The budget changed after B looked. B is told why, and can look again.

IN PLAIN WORDS

Nothing is overspent, and the second helper is told exactly why it was stopped.

UNDER THE HOOD

Authorization is evaluated against the version the agent read; success consumes that version, so a stale read can never become a write.

WHAT KALDRIX DOES

Four powers. One desk.



Always current

Acts only on what is still true. If the world moved while an agent was thinking, it is told to look again.

VERSION-CHECKED EXECUTION



Shared money, never overspent

A group of agents shares one allowance. However many ask at once, the total can never be exceeded.

RESERVE-THEN-CAPTURE SPEND LEDGER



Permissions that can only shrink

Signed, limited, expiring and cancellable on the spot. A helper can never hold more than the agent that made it.

ATTENUATING CAPABILITY CHAINS · ED25519 EXECUTION PERMITS



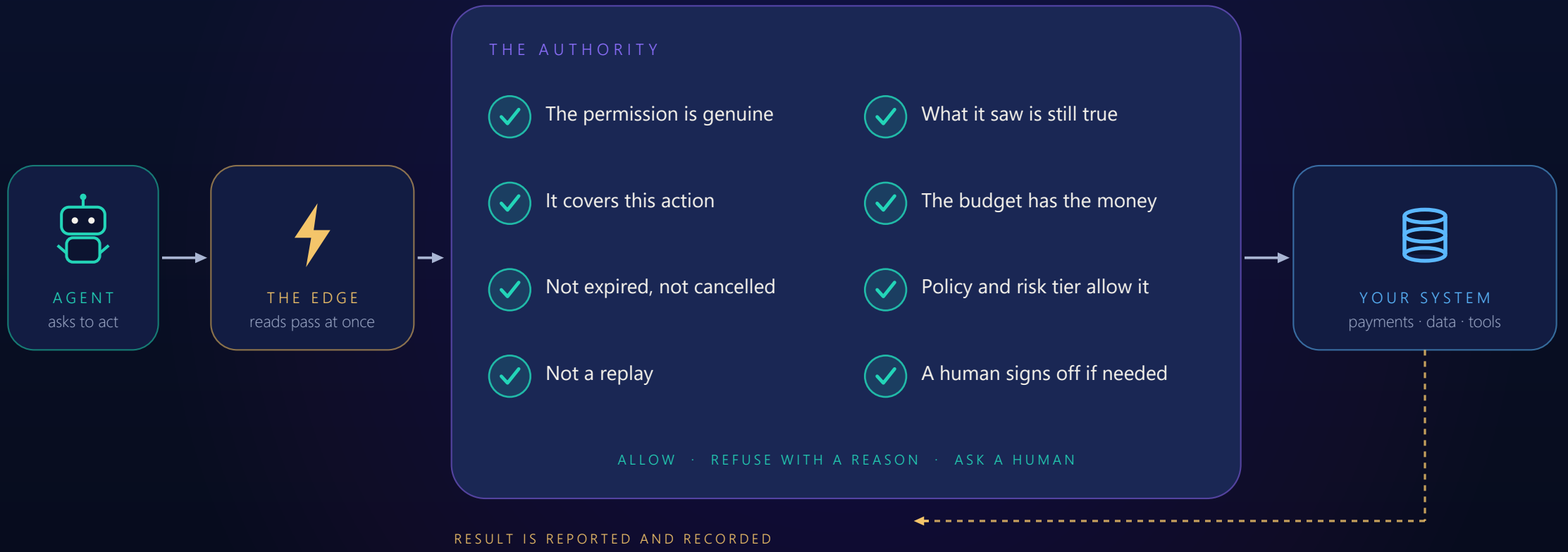
A record nobody can quietly rewrite

Every decision is written down with its reason, linked to the last so any edit shows.

SHA-256 HASH-CHAINED AUDIT LOG

INSIDE A DECISION

Eight checks, one answer, in a fraction of a millisecond.



IN PLAIN WORDS

Every request is checked eight ways, then answered with a yes, a no and the reason, or a call for a human.

UNDER THE HOOD

Median authorize 56 µs · full purchase path 247 µs · reads on the edge fast path 1.5 µs (measured, one laptop process).

PERMISSIONS

Authority can be handed down. It can never grow.



Narrow, never widen

A copy can only hold less than the permission it came from.

Cancel one, cancel all beneath

Pull a permission and every copy handed down from it stops at once.

Signed and personal

Each permission names its holder, expires on its own, and cannot be edited.

IN PLAIN WORDS

Give a team a budget and every agent below it gets only a slice. Nobody can quietly give themselves more.

UNDER THE HOOD

Macaroon-style HMAC-SHA256 chains with attenuation-only caveats; revocation propagates to every delegate and survives restart.

SHARED MONEY

Money is held before it is spent.

ALL-OR-NOTHING ACROSS THE WHOLE CHAIN OF DELEGATION



A team allowance of $50,000

1 Reserve

The moment an agent asks, the money is set aside. Nobody else can count it.

2 Capture

When the action succeeds, the hold becomes a real, final spend.

3 Release

If it fails or times out, the hold returns to the pool for others to use.

IN PLAIN WORDS

Like a hotel putting a hold on a card: the money is promised to one agent before the next one is served.

UNDER THE HOOD

Two-phase spend accounting over a shared ledger, checked under every ordering of simultaneous requests by an exhaustive model checker.

AT SCALE

The same desk, from one agent to a whole network.



One agent

A single helper doing a single job.

- Always current ✓
- Shared money ✓
- Shrinking permissions ✓
- Permanent record ✓



A team

Helpers sharing one budget.

- Always current ✓
- Shared money ✓
- Shrinking permissions ✓
- Permanent record ✓



A fleet

Departments of agents running whole processes.

- Always current ✓
- Shared money ✓
- Shrinking permissions ✓
- Permanent record ✓



A network

Agents from different companies dealing with each other.

- Always current ✓
- Shared money ✓
- Shrinking permissions ✓
- Permanent record ✓

IN PLAIN WORDS

Whether it is one helper or many, the same desk applies the same rules, and every function keeps working.

UNDER THE HOOD

Verification is stateless signed-token checking; shared state lives only in the spend ledger and the audit log. Tested under 16 and 24 simultaneous payments.

THE RECORD

A record nobody can quietly rewrite.



What is written

Who asked, what they asked for, the verdict and the reason, and the time. For every decision, allowed or refused.

How it is protected

Each record carries a fingerprint of the one before it. The log can be added to, never edited in place.

How you check

One call recomputes the whole chain. If anything was touched, it says exactly where.

IN PLAIN WORDS

Every decision is written down with its reason. If anyone changes the story afterwards, the change shows.

UNDER THE HOOD

Append-only log (the database refuses edits and deletes); SHA-256 hash chain; one call recomputes the chain and returns the head to store elsewhere.

SPEED

Fast enough to sit in front of every single action.



LOGARITHMIC SCALE: EACH STEP RIGHT IS TEN TIMES LONGER

131 µs
median time for a full decision: chain, holder, proof, replay and scope

≈ 7,000
full decisions per second from one thread of one laptop

0
wrong answers across 20,000 timed decisions in every test row

IN PLAIN WORDS

The check takes a tiny fraction of a blink, so putting it in front of every action costs nothing you would notice.

UNDER THE HOOD

Ed25519 strict verification dominates the cost (about 65 µs per delegation block). Measured 17 Oct 2026, release build, one Windows laptop, one caller at a time.

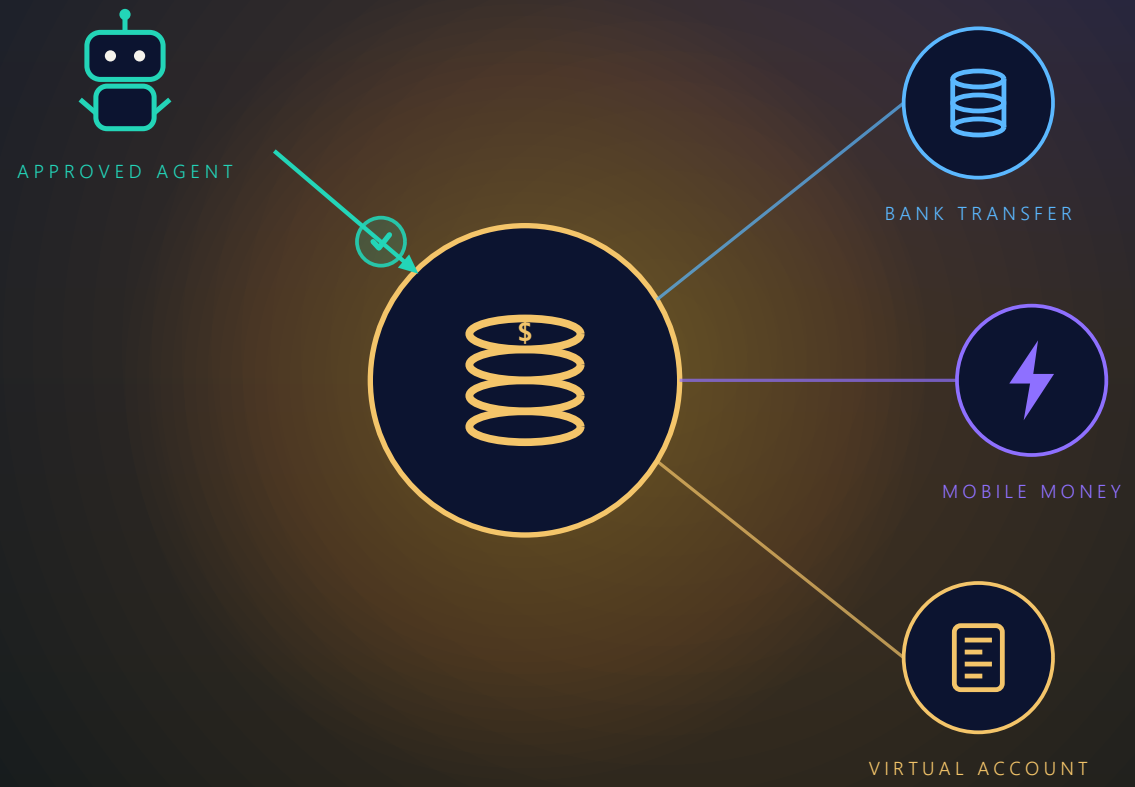
PART TWO · AGENTIC LIQUIDITY AND LOCAL-RAIL GATEWAY

Zentrix

Where AI agents move money: safely, locally, globally.

Kaldrix decides what an agent may do. Zentrix carries it out on the payment rails people actually use.

BUILT FOR GLOBAL USE | NIGERIA FIRST | PLUGS INTO ANY AUTHORITY



ONE GATEWAY, MANY LOCAL RAILS

WHAT ZENTRIX DOES

Four promises for every payment.



Pays on local rails

Bank transfer, mobile money and virtual accounts, country by country. A new market is a new adapter, not a rewrite.

CORRIDOR = COUNTRY · CURRENCY · RAIL



Pays exactly what was approved

The approval names the amount, the recipient and the purpose. Change any of it and the payment stops.

SIGNED CONSTRAINTS · TRANSACTION DIGEST



Never pays twice

Retry as often as you like. The same request always returns the same payment, and one approval pays once.

IDEMPOTENCY · ATOMIC LIMITS · INTEGER MINOR UNITS



Always explainable

Every payment has a status you can trust, a receipt, and a trail back to who approved it.

STATE MACHINE · RECEIPTS · HASH-CHAINED AUDIT

IN PLAIN WORDS

Money moves only when it is allowed, only as far as it was allowed, and only once.

UNDER THE HOOD

Amounts are integers in each currency's minor unit; limits are checked atomically with the state change; callbacks are idempotent; receipts come from stored data.

INSIDE A PAYMENT

Seven stages. Every one on the record.

1

Request received

The instruction arrives with its own idempotency key.

2

Request validated

Well formed, amounts in whole minor units, key claimed.

3

Authorization checked

Kaldrix, or your own authority, must say yes to this exact payment.

THE SECURITY GATE

4

Payment rules checked

Limits, approved recipients and the corridor are enforced.

5

Sent to the provider

Only now does a provider see it. Accepted is not settled.

6

Provider reports back

Success, failure or unknown, reported as it is.

7

Final status confirmed

Settlement is confirmed and a receipt is produced.

REFUSED AT ANY STAGE

It stops there, with a reason on the record. The payment provider never sees it.

IN PLAIN WORDS

A payment is a short, visible journey. At any point you can see where it is and why it stopped.

UNDER THE HOOD

Each stage is derived only from events persisted for that payment; a stage is shown as done only if an event says so.

EXACT-PAYMENT BINDING

Approved for this payment. Not for any other.

THE SIGNED APPROVAL

AMOUNT
$25,000.00

RECIPIENT
Ada O.

PURPOSE
Supplier payment

REFERENCE
inv-1042

SEALED WITH 7c1e…b94f

PAYMENT A · SAME AS APPROVED

$25,000.00 to Ada O. for supplier payment

Fingerprint 7c1e…b94f matches the seal

PAID, ONCE

PAYMENT B · ONE DETAIL CHANGED

$250,000.00 to Ada O. for supplier payment

Fingerprint e03a…1d72 does not match the seal

REFUSED · authorization_binding

IN PLAIN WORDS

The approval is for one exact payment. A stolen or altered approval cannot be used to pay anything else.

UNDER THE HOOD

A SHA-256 digest of the payment plus signed amount, currency, recipient and purpose; a mismatch on any gets its own rejection code.

BRING ANY AUTHORITY

Zentrix trusts only the checks it can see were done.

Kaldrix

Signed, exact-payment decisions. Verified end to end.

Another platform

Plug in its adapter. It declares what it can check.

Your own service

Use the approval system you already run.



EACH PROVIDER DECLARES WHAT IT CAN CHECK

- ✓ Who signed it
- ✓ Has it expired
- ✓ Has it been revoked
- ✓ Is the amount allowed
- ✓ Is the currency allowed
- ✓ Is the recipient allowed
- ✓ Is the purpose allowed
- ✓ Is it this exact payment

A yes that skipped a check the account requires is treated as a no. A provider that cannot be reached, or gives no clear answer, is a no.

FAIL CLOSED

NEVER CLAIMS A CHECK IT DID NOT DO

IN PLAIN WORDS

Zentrix does not force one security system on you. It works with yours, and says plainly what each one checked.

UNDER THE HOOD

Declared capabilities and performed checks per provider; required checks per account; the Kaldrix adapter verifies Ed25519 signatures with public keys only.

GLOBAL BY DESIGN

Built for the world. Starting where it matters first: Nigeria.

NIGERIA FIRST



12 CURRENCIES CARRIED BY THE ENGINE, EACH WITH ITS OWN DECIMALS AND LIMITS

NGN · GHS · KES · ZAR · EGP · TZS · UGX · XOF · XAF · USD · EUR · GBP

HOW A MARKET IS ADDED

1. **Add the payment provider's adapter**
   One adapter per provider: the bank, the mobile-money operator, the local rail.
2. **Declare its corridor**
   A corridor is a country, a currency and a rail.
3. **Set limits and switch it on**
   Per currency, per account. Nothing is on by default.

THE PAYMENT ENGINE ITSELF NEVER CHANGES.

IN PLAIN WORDS

A new country is a new adapter and a setting, not a rebuild. The safety rules travel with every corridor.

UNDER THE HOOD

Country · currency · rail corridors; per-currency limits; routing by declared corridor. A corridor with no provider is refused, never rerouted.

SAFE BY CONSTRUCTION

Six habits that keep money from going wrong.



Whole-number money

Amounts are whole numbers of the smallest unit, so rounding errors cannot happen.



Same request, same payment

A retry returns the original. Reusing a key for a different payment is refused.



Limits that hold under pressure

Single-payment and daily limits are enforced atomically, tested with 24 payments at once.



Honest about the unknown

If a provider does not answer, the payment is marked unknown: never guessed paid or failed.



Reconciliation

Every recorded payment can be compared with the provider's own statement, and any difference shown.



Receipts from stored facts

A receipt is generated from what was recorded, never from what a caller claims.

IN PLAIN WORDS

Money software fails in small, boring ways. Each of these habits closes one of them.

UNDER THE HOOD

Integer minor units; unique idempotency keys; atomic limit checks; a validated state machine with an unknown-outcome state; receipts from stored rows.

THE OPPORTUNITY

Agents are about to move money. Who is in charge?

TODAY

Agents act



They book, buy, update and send on their own.

NEXT

Agents pay



The same agents settle invoices, run payouts and move funds.

ALWAYS

Someone must answer



For every payment, a person will be asked who allowed it.

THE QUESTION EVERY BANK, REGULATOR AND FINANCE TEAM WILL ASK

Who approved this payment, for how much, to whom, and can you prove it?

IN PLAIN WORDS

The more agents handle money, the more this question matters. Kaldrix and Zentrix answer it by design, on every payment.

UNDER THE HOOD

Signed decision bound to the exact payment, a verified chain of authority, and a tamper-evident record: the evidence exists before anyone asks for it.

WHO IT IS FOR

Anyone who lets software spend money.



Fintechs and payment apps

Let customers' AI assistants pay bills and suppliers, inside limits the customer set.

PER-CUSTOMER LIMITS



Payroll and payouts

Agents run payout batches under a budget. Nothing leaves that was not approved.

BUDGETS · APPROVALS



Marketplaces and platforms

Pay sellers, drivers and creators automatically, with a full trail for every payment.

AUTOMATIC PAYOUTS · FULL TRAIL



Enterprises running agent fleets

Give every agent its own budget and audit trail, controlled from one desk.

FLEET-WIDE CONTROL

IN PLAIN WORDS

If an AI can spend it, someone has to set the limits, check each payment and keep the proof. That is the job.

UNDER THE HOOD

Delegated budgets, exact-payment approval and an audit trail, exposed as an HTTP API with a TypeScript SDK; the authority is pluggable.

THE CONTROL ROOM

See every decision, as it happens.



LAYOUT OF THE CONTROL ROOM · FIGURES SHOWN ARE ILLUSTRATIVE

Two guided labs

Seven security scenarios and seven payment scenarios, run against the live system with the real result shown.

Live stream

Watch decisions arrive as they are made.

Audit and verification

Search every decision and recompute the chain on demand.

Security and authorization

Which authority is configured, and exactly what it can verify.

IN PLAIN WORDS

Everything the system decides is visible in one place, in plain language, with the reason beside it.

UNDER THE HOOD

React single-page app served by the platform; live stream over server-sent events with single-use tickets; every figure is counted from persisted events.

SEE FOR YOURSELF

Do not take our word for it. Open it.

SCAN, OR VISIT

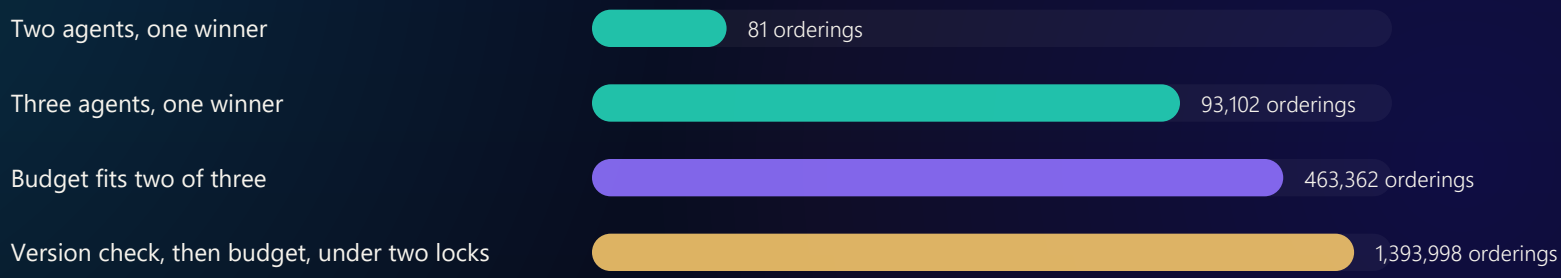
kaldrix-proof.vercel.app

A RECORDED RUN, STEP BY STEP

Five situations that go wrong when software agents handle money. Each is a real run of the real system. Step through and watch what Kaldrix decides.

THE HARD CASE: TWO AGENTS AT THE SAME INSTANT

The checks that carry the promise were run under a tool that tries every possible ordering of the threads.



HELD IN EVERY ONE. THE CHECKER WAS ALSO SHOWN IT CAN FAIL, SO A GREEN RESULT MEANS SOMETHING.

IN PLAIN WORDS

Open the link on any phone or laptop and follow the system making its decisions, one step at a time.

UNDER THE HOOD

Recorded with a fixed clock and fixed randomness, so re-running gives the identical recording; concurrency claims backed by exhaustive interleaving checks.

BUILT, MEASURED, RUNNING

What exists today.

894
automated tests passing across the platform

17
required payment-safety tests, all passing

7 + 7
security and payment scenarios you can run live

131 µs
median time for a full authorization decision

12
currencies carried, each exact to its smallest unit

3 of 3
high-severity findings from a separate security review, fixed and re-tested

WHAT COMES NEXT

More corridors
Ghana, Kenya and South Africa

A production database
PostgreSQL behind the same store

Live provider connections
Real banks and mobile-money operators

IN PLAIN WORDS

The foundation is built and tested. What remains is adding places for it to run and money rails for it to reach.

UNDER THE HOOD

Rust workspace, SQLite store with a documented PostgreSQL path, React control room, TypeScript SDK, CI for the Rust, web and SDK packages.

KALDRIX · ZENTRIX

Every agent.
Every action.
Every payment.

Decided by Kaldrix. Moved by Zentrix. Remembered for good.



CHECKED PROTECTED REMEMBERED

SEE IT RUN · kaldrix-proof.vercel.app